Lu's Three-Chord Identity L3 and the Cyclic Identity CL3

Distance Relations Among Six Points on a Circle, Three Chords, and Their Triangle of Intersections: Rigorous Proofs, Structural Comparisons, and Generalizations

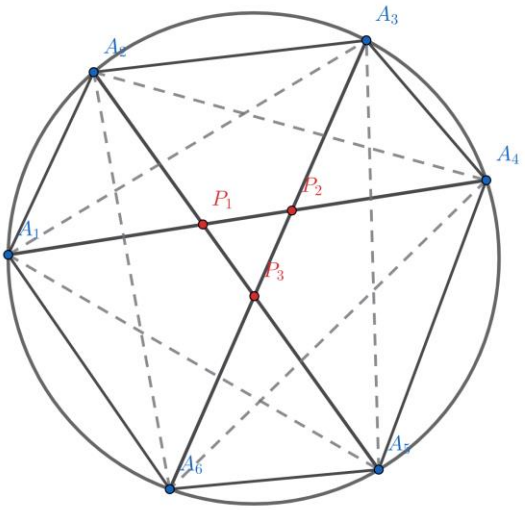
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Manuscript note: This paper retains the agreed notation $A_1,\ldots,A_6$ and P_1,P_2,P_3. The main text provides a complete trigonometric-identity proof of L3, as well as a proof of CL3 obtained by summing over six cyclic shifts and collecting coefficients term by term.

Abstract

Let $A_1,\ldots,A_6$ be six distinct points on the same circle, arranged in their cyclic order. Draw the three chords $\ell_1=A_1A_4$, $\ell_2=A_2A_5$, and $\ell_3=A_3A_6$, and define $P_1=\ell_2\cap\ell_3$, $P_2=\ell_1\cap\ell_3$, and $P_3=\ell_1\cap\ell_2$. This paper studies Lu's three-chord identity L3 and the cyclic-collection identity CL3. It presents an analytic proof of L3 and derives CL3 by cyclic summation. Their configurations, algebraic forms, and geometric roles are compared with the intersecting chords theorem, Ptolemy's theorem, Fuhrmann's theorem, McDougall's even-point generalization, and Ceva's theorem for chords. The paper explicitly distinguishes the configuration in which the three chords intersect pairwise to form a triangle of intersection points from the configuration in which the three chords are concurrent. The comparison with classical results is intended to clarify the mathematical background and structural differences; it does not, by itself, establish irreducibility or historical priority.

Keywords: six points on a circle; Lu's three-chord identity; cyclic identity; intersecting chords theorem; Ptolemy's theorem; Fuhrmann's theorem; McDougall's theorem; Ceva's theorem for chords



$$A_1A_2 \cdot A_1A_5 \cdot P_1P_3 + A_2A_3 \cdot A_2A_6 \cdot P_1P_2 = A_1A_2 \cdot A_2A_4 \cdot P_2P_3 + A_1A_3 \cdot A_1A_6 \cdot P_1P_2$$

The identity above holds (L3).

1. Introduction

Classical metric relations in circle geometry usually concern the products of segments formed at a single chord intersection, the sides and diagonals of a cyclic quadrilateral, or the relationship between chord lengths and inscribed angles. This paper focuses on a configuration involving three chords simultaneously: the three chords intersect pairwise, and their three intersection points form a triangle. The target equation directly relates the chord lengths determined by the original six points to the side lengths of the triangle of intersection points.

This paper distinguishes three questions: first, whether the formula holds in the specified configuration; second, whether it admits a rigorous proof whose steps can be checked individually; and third, whether the identity has appeared in the existing literature or can be derived by a simple combination of classical theorems. Analytic proofs and computational cross-checks are provided for the first two questions. For the third, the paper offers only a structural comparison and does not equate differences in form with mathematical independence or historical originality.

2. Configuration, Notation, and Nondegeneracy Conditions

Let A_1,A_2,A_3,A_4,A_5,A_6 be six distinct points on the same circle, arranged in their cyclic order. Draw the three chords:

$$\ell_1 = A_1A_4, \quad \ell_2 = A_2A_5, \quad \ell_3 = A_3A_6.$$

The intersection points are defined according to the fixed diagram and are not arbitrarily renamed during the cyclic derivation:

$$P_1 = A_2A_5 \cap A_3A_6; \quad P_2 = A_1A_4 \cap A_3A_6; \quad P_3 = A_1A_4 \cap A_2A_5.$$

The notation A_iA_j denotes the Euclidean distance $|A_iA_j|$, and P_iP_j denotes the Euclidean distance between the intersection points. Since the six points are strictly ordered around the circle, all three chords are interior chords and intersect pairwise. The main proof is given for the nondegenerate case in which the triangle of intersection points is nondegenerate. If the three chords are concurrent, then $P_1=P_2=P_3$, all terms on both sides of L3 and CL3 are zero, and the identities hold trivially.

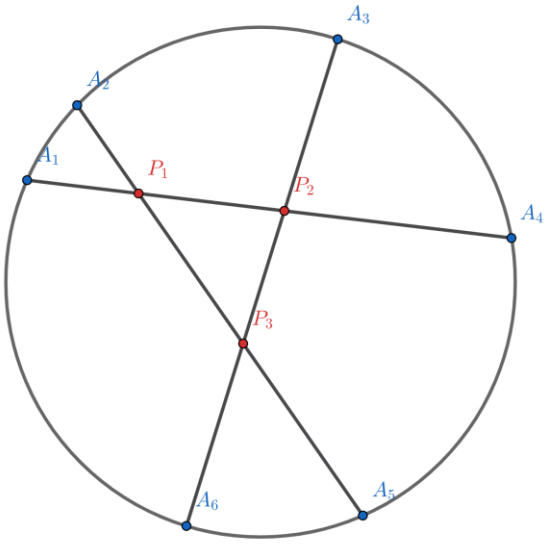


Figure 1. The circle with six points, three chords, and intersection points P_1, P_2, P_3 labeled according to the fixed definitions.

3. Lu's Three-Chord Identity L3

Theorem 3.1 (L3). In the configuration of Section 2, the following identity always holds:

$$A_1A_2 \cdot A_1A_5 \cdot P_1P_3 + A_2A_3 \cdot A_2A_6 \cdot P_1P_2 = A_1A_2 \cdot A_2A_4 \cdot P_2P_3 + A_1A_3 \cdot A_1A_6 \cdot P_1P_2 \qquad \text{(L3)}$$

Each term in L3 is a product of three lengths; hence the identity is homogeneous of degree three. Under a uniform similarity scaling by λ, every length is multiplied by λ, so both sides of the identity are multiplied by λ^3.

4. Rigorous Analytic Proof of L3

4.1 Parameterization by Half-Central Angles

Let the radius of the circle be R. Choose half-central-angle parameters increasing along the circumference, $u_1<\ldots<u_6<u_1+\pi$, and set

$$\mathbf{a=u_2-u_1,\quad b=u_3-u_2,\quad c=u_4-u_3,\quad d=u_5-u_4,\quad e=u_6-u_5.}$$

Thus a,b,c,d,e>0 and $a+b+c+d+e<\pi$. Let point A_i correspond to the central angle $2u_i$. Since all relevant angle sums lie in $(0,\pi)$, the sines in the following chord-length formulas are positive:

$A_1A_2 = 2R\sin a$, $A_1A_5 = 2R\sin(a+b+c+d)$,

$A_2A_3 = 2R\sin b$, $A_2A_6 = 2R\sin(b+c+d+e)$,

$A_2A_4 = 2R\sin(b+c)$, $A_1A_3 = 2R\sin(a+b)$,

$A_1A_6 = 2R\sin(a+b+c+d+e)$。

4.2 The Three Sides of the Triangle of Intersection Points

The direction angle of chord A_iA_j, modulo π, is $u_i+u_j+\pi/2$. Therefore, the angle between the lines containing two chords is the absolute value of the difference between their corresponding direction angles. Let ρ be the circumradius of triangle $P_1P_2P_3$. By the extended law of sines, a side length equals 2ρ times the sine of its opposite angle. Computing the differences among the direction angles of the three pairs of chords gives:

$$P_1P_2 = 2\rho \sin(a+d),\quad P_1P_3 = 2\rho \sin(a+b+d+e),\quad P_2P_3 = 2\rho \sin(b+e).$$

For example, P_1P_2 lies on chord A_3A_6, and its opposite angle is the angle between the lines A_1A_4 and A_2A_5 at P_3. The difference between the direction angles of these two chords is $(u_1+u_4)-(u_2+u_5)=-(a+d)$, so the sine of that angle is $\sin(a+d)$. The other two formulas follow in the same way, using the angle between A_1A_4 and A_3A_6 at P_2, and the angle between A_2A_5 and A_3A_6 at P_1, respectively. Since $a+d$, $a+b+d+e$, and $b+e$ all belong to $(0,\pi)$, these sines are positive.

4.3 Substitution and Reduction to a Trigonometric Identity

Substituting the expressions from Sections 4.1 and 4.2 into L3, all four terms contain the common factor $8R^2\rho$. Dividing by this positive factor, L3 is equivalent to

$$\sin a\cdot\sin(a+b+c+d)\cdot\sin(a+b+d+e)+\sin b\,\sin(b+c+d+e)\,\sin(a+d)$$
$$=\sin a\cdot\sin(b+c)\cdot\sin(b+e)+\sin(a+b)\,\sin(a+b+c+d+e)\,\sin(a+d). \quad (1)$$

Let $K=a+2b+c+d+e$. Using the product-to-sum formula $2\sin X\sin Y=\cos(X-Y)-\cos(X+Y)$, we obtain the following two auxiliary identities.

4.4 Step-by-Step Proofs of Two Auxiliary Identities

Lemma 4.1.

$$\sin b\cdot\sin(b+c+d+e)-\sin(a+b)\cdot\sin(a+b+c+d+e)=-\sin a\cdot\sin K. \quad (2)$$

Proof. Multiply the left-hand side by 2 and apply the product-to-sum formula to each term:

$$2\sin b\sin(b+c+d+e)=\cos(c+d+e)-\cos(2b+c+d+e),$$

$$2\sin(a+b)\sin(a+b+c+d+e)=\cos(c+d+e)-\cos(2a+2b+c+d+e).$$

Subtracting, the two terms $\cos(c+d+e)$ on the left cancel, leaving

$$\cos(2a+2b+c+d+e)-\cos(2b+c+d+e)=-2\sin(a+2b+c+d+e)\sin a=-2\sin K\sin a.$$

Dividing both sides by 2 gives equation (2). (Q.E.D).

Lemma 4.2.

$$\sin(a+b+c+d)\cdot\sin(a+b+d+e)-\sin(b+c)\cdot\sin(b+e)=\sin(a+d)\cdot\sin K. \quad (3)$$

Proof. Again, use the product-to-sum formula:

$2\sin(a+b+c+d)\sin(a+b+d+e) = \cos(c-e) - \cos(2a+2b+c+2d+e)$,

$2\sin(b+c)\sin(b+e) = \cos(c-e) - \cos(2b+c+e)$。

Subtracting the two equations, the terms $\cos(c-e)$ cancel, leaving

$\cos(2b+c+e) - \cos(2a+2b+c+2d+e) = 2\sin(a+d)\sin(a+2b+c+d+e) = 2\sin(a+d)\sin K$。

Dividing both sides by 2 gives equation (3). (Q.E.D).

4.5 Completion of the Proof of L3

Subtract the right-hand side of equation (1) from its left-hand side, and group the terms by sin a and sin(a+d). The difference F can be written as:

$F = \sin a\cdot[\sin(a+b+c+d)\sin(a+b+d+e) - \sin(b+c)\sin(b+e)]$

$+ \sin(a+d)\cdot[\sin b\sin(b+c+d+e) - \sin(a+b)\sin(a+b+c+d+e)]$。

By Lemma 4.2, the first bracket equals $\sin(a+d)\sin K$; by Lemma 4.1, the second bracket equals $-\sin a\sin K$. Therefore,

$F = \sin a\cdot\sin(a+d)\cdot\sin K - \sin(a+d)\cdot\sin a\cdot\sin K = 0$。

Thus equation (1) holds, and consequently L3 holds. (Q.E.D).

5. The Cyclic Lu Identity CL3

Theorem 5.1 (CL3). In the same configuration, the following cyclic-collection identity holds:

$$(A_1A_2\cdot A_1A_5 + A_4A_5\cdot A_2A_4)\cdot P_1P_3 + (A_1A_3\cdot A_3A_4 + A_1A_6\cdot A_4A_6)\cdot P_2P_3 + (A_2A_3\cdot A_2A_6 + A_5A_6\cdot A_3A_5)\cdot P_1P_2$$
$$= (A_1A_2\cdot A_2A_4 + A_1A_5\cdot A_4A_5)\cdot P_2P_3$$
$$+ (A_1A_3\cdot A_1A_6 + A_3A_4\cdot A_4A_6)\cdot P_1P_2$$
$$+ (A_2A_3\cdot A_3A_5 + A_5A_6\cdot A_2A_6)\cdot P_1P_3。 \quad \text{(CL3)}$$

CL3 is not obtained merely by mechanically replacing the fixed intersection-point names cyclically. A rigorous derivation requires redefining the local intersection points for each cyclic shift and then mapping them back to the fixed points P_1, P_2, and P_3. The complete six-row audit is given below.

6. Rigorous Proof of CL3 by Cyclic Summation

For k=0,1,…,5, let B_i=A_{i+k}, with subscripts taken modulo 6 and represented by 1,…,6. For each k, define the local chords $m_1=B_1B_4$, $m_2=B_2B_5$, and $m_3=B_3B_6$, together with $Q_1=m_2\cap m_3$, $Q_2=m_1\cap m_3$, and $Q_3=m_1\cap m_2$. By the already-proved identity L3, the six shifts respectively satisfy

$$B_1B_2\cdot B_1B_5\cdot Q_1Q_3 + B_2B_3\cdot B_2B_6\cdot Q_1Q_2 = B_1B_2\cdot B_2B_4\cdot Q_2Q_3 + B_1B_3\cdot B_1B_6\cdot Q_1Q_2。 \quad (L3_k)$$

Because the three chords after each shift are still the original ℓ_1, ℓ_2, and ℓ_3, only in a cyclically changed order, the local intersection points must be the fixed points P_1, P_2, and P_3. The correspondence is as follows.

Shift k	Local point order $B_1,\ldots,B_6$	Local chords (m_1,m_2,m_3)	(Q_1,Q_2,Q_3)	(Q_1Q_3,Q_1Q_2,Q_2Q_3)
0	A_1,A_2,A_3,A_4,A_5,A_6	(ℓ_1,ℓ_2,ℓ_3)	(P_1,P_2,P_3)	(P_1P_3,P_1P_2,P_2P_3)
1	A_2,A_3,A_4,A_5,A_6,A_1	(ℓ_2,ℓ_3,ℓ_1)	(P_2,P_3,P_1)	(P_1P_2,P_2P_3,P_1P_3)
2	A_3,A_4,A_5,A_6,A_1,A_2	(ℓ_3,ℓ_1,ℓ_2)	(P_3,P_1,P_2)	(P_2P_3,P_1P_3,P_1P_2)
3	A_4,A_5,A_6,A_1,A_2,A_3	(ℓ_1,ℓ_2,ℓ_3)	(P_1,P_2,P_3)	(P_1P_3,P_1P_2,P_2P_3)
4	A_5,A_6,A_1,A_2,A_3,A_4	(ℓ_2,ℓ_3,ℓ_1)	(P_2,P_3,P_1)	(P_1P_2,P_2P_3,P_1P_3)
5	A_6,A_1,A_2,A_3,A_4,A_5	(ℓ_3,ℓ_1,ℓ_2)	(P_3,P_1,P_2)	(P_2P_3,P_1P_3,P_1P_2)

When collecting terms from the six equations, all coefficients are products of chord lengths. To make multiplicities explicit in the long expressions, the terms in which each fixed side between intersection points occurs on the left and right of the six equations are listed below. Each entry occurs twice, so every coefficient in the sum is twice the sum of the corresponding two terms.

6.1 Coefficients on the Left-Hand Side

- Coefficient of P_1P_3 on the left: $A_1A_2 \cdot A_1A_5$ occurs twice, and $A_4A_5 \cdot A_2A_4$ occurs twice; the total coefficient is $2(A_1A_2 \cdot A_1A_5 + A_4A_5 \cdot A_2A_4)$.
- Coefficient of P_2P_3 on the left: $A_1A_3 \cdot A_3A_4$ occurs twice, and $A_1A_6 \cdot A_4A_6$ occurs twice; the total coefficient is $2(A_1A_3 \cdot A_3A_4 + A_1A_6 \cdot A_4A_6)$.
- Coefficient of P_1P_2 on the left: $A_2A_3 \cdot A_2A_6$ occurs twice, and $A_5A_6 \cdot A_3A_5$ occurs twice; the total coefficient is $2(A_2A_3 \cdot A_2A_6 + A_5A_6 \cdot A_3A_5)$.

6.2 Coefficients on the Right-Hand Side

- Coefficient of P_2P_3 on the right: $A_1A_2 \cdot A_2A_4$ occurs twice, and $A_1A_5 \cdot A_4A_5$ occurs twice; the total coefficient is $2(A_1A_2 \cdot A_2A_4 + A_1A_5 \cdot A_4A_5)$.
- Coefficient of P_1P_2 on the right: $A_1A_3 \cdot A_1A_6$ occurs twice, and $A_3A_4 \cdot A_4A_6$ occurs twice; the total coefficient is $2(A_1A_3 \cdot A_1A_6 + A_3A_4 \cdot A_4A_6)$.
- Coefficient of P_1P_3 on the right: $A_2A_3 \cdot A_3A_5$ occurs twice, and $A_5A_6 \cdot A_2A_6$ occurs twice; the total coefficient is $2(A_2A_3 \cdot A_3A_5 + A_5A_6 \cdot A_2A_6)$.

Adding the six equations $L3_k$, the coefficient of each fixed side between intersection points on both sides contains the common factor 2. Dividing both sides by 2 gives exactly CL3 in Theorem 5.1. This process proves both the validity of CL3 and its cyclic origin. Q.E.D.

7. Cyclic Symmetry and Homogeneity

Cyclic structure. L3 holds for any six points labeled in their cyclic order. Under each of the six cyclic shifts $A_i \mapsto A_{i+1}$, the local chords and local intersection points must be redefined together. CL3 is the averaged form of the six shifted identities (the sum of the six equations divided by 2), and therefore has a structure organized by the cyclic order of the six points. Here, “cyclic symmetry” means that the relation as a whole is preserved under cyclic relabeling, not that every individual term remains unchanged under the fixed labels.

Homogeneity. Every term in L3 and CL3 is a product of three length factors. If all points undergo a similarity scaling $x \mapsto \lambda x$ ($\lambda>0$), then every chord length and every distance between intersection

points is multiplied by λ, so both sides are multiplied by λ^3. The expressions in parentheses are homogeneous of degree two; multiplying them by a side of the triangle of intersection points gives degree-three homogeneity.

Reflection and relabeling. Reversing the direction in which the points are read around the circle induces a reversal of labels. To write an explicit reflected form, the labels of all intersection points must be recalculated and the mapping of every term checked. One cannot conclude, solely from the appearance of the diagram, that the formula is termwise invariant under the fixed labels.

8. Structural Comparison with Five Classical Theorems and Generalizations

8.1 Intersecting Chords Theorem

Intersecting chords theorem: If chords AB and CD intersect at an interior point X of a circle, then $XA \cdot XB = XC \cdot XD$. Its structure is a local product relation: the products of the four segments into which two chords are divided at a single intersection point are equal. By contrast, L3/CL3 treats the three chords as a whole, uses the side lengths of the triangle formed by their three pairwise intersection points, and takes products of chord lengths between the original six points as coefficients. The objects and organization of terms differ; however, this structural difference alone does not prove that L3 cannot be derived by algebraic elimination from several applications of the intersecting chords theorem. The proof of L3 given here uses chord direction angles, the extended law of sines, and trigonometric identities independently, without relying on the intersecting chords theorem as an intermediate step.

8.2 Ptolemy's Theorem

Ptolemy's theorem: For a cyclic quadrilateral ABCD, $AB \cdot CD + BC \cdot DA = AC \cdot BD$. Its central statement is that the sum of the products of the two pairs of opposite sides equals the product of the diagonals. L3 involves six points on a circle, three specified chords, and the distances between their three intersection points; CL3 groups products of chord lengths and multiplies them by the side lengths of the triangle of intersection points. The two results operate at different levels of combination, and L3 is not a direct same-form restatement of Ptolemy's formula. To claim further that it cannot be derived by a simple combination of Ptolemy relations would require systematic algebraic elimination among the possible four-point subconfigurations. This paper does not exaggerate "not a direct restatement" into "algebraic independence has been proved."

Comparison dimension	Intersecting Chords Theorem	Ptolemy's Theorem	L3 / CL3
Basic number of points	Four endpoints on the circle and one intersection	Four points on the circle	Six points on the circle and three chord intersections
Central objects	Four chord segments at the same intersection	Sides and diagonals of a quadrilateral	Original chord lengths and sides of the triangle of intersection points

Comparison dimension	Intersecting Chords Theorem	Ptolemy's Theorem	L3 / CL3
Typical form	Length × length = length × length	Sum of two products = one product	Sum of two products of three lengths = two products of three lengths
Homogeneous degree	2	2	3
Organization	Local intersection-point product	Opposite sides/diagonals of a quadrilateral	Coupling of three chords; cyclic collection in CL3
Theorem/relation	Basic configuration	Type of central relation	Difference from L3/CL3
Intersecting Chords Theorem	Two chords intersect at one point	Equality of two products of segment lengths	Local four chord-segment lengths; L3/CL3 involve distances among three intersection points
Ptolemy's Theorem	Cyclic quadrilateral	Sum of products of two pairs of opposite sides equals the product of the diagonals	Four points, degree-two homogeneity; L3/CL3 involve six points and degree-three homogeneity
Fuhrmann's Theorem	Convex cyclic hexagon	Cubic relation among side lengths and three specified diagonals	Side–diagonal relation; L3/CL3 introduce the side lengths of the triangle of intersection points
McDougall relation	Any 2n points on a circle	Alternating sum of reciprocals of global distance products	Reciprocal sum for an even number of points; L3/CL3 concern the triangle formed by three chord intersections
Ceva's Theorem for Chords	Six-point circle and three alternately paired chords	Necessary and sufficient product criterion for concurrency of the three chords	Describes the degenerate concurrent case; the main subject of L3/CL3 is a nondegenerate triangle of intersection points
L3 / CL3	Six-point circle; three chords intersect pairwise	Cubic homogeneous identity involving products of chord lengths and side lengths of the triangle of intersection points	Subject of this paper

8.3 Fuhrmann's Theorem

Fuhrmann's theorem is a classical generalization of Ptolemy's relation to a convex cyclic hexagon. If the opposite sides of the hexagon are labeled in the standard way as a,a′,b,b′,c,c′, and three diagonals e,f,g satisfying the theorem's labeling conditions are selected, then efg = aa′e + bb′f + cc′g + abc + a′b′c′. This is a homogeneous identity of degree three involving side lengths and diagonal lengths. L3/CL3 are also homogeneous of degree three, but their key variables additionally include the distances between the pairwise intersections of three specified chords. The geometric objects and organization of terms are different.

8.4 McDougall's Even-Point Generalization

Let $P_1,\ldots,P_{2n}$ be 2n points arranged in order on a circle, and define $R_i=\prod_{j\neq i}|P_iP_j|$; that is, R_i is the product of the distances from P_i to all the other points on the circle. The McDougall relation is

$\sum_{i=1}^{n} 1/R_{2i} = \sum_{i=1}^{n} 1/R_{2i-1}$. This relation extends a Ptolemy-type relation to any even number of points on a circle and uses an alternating sum of reciprocals of global distance products. By contrast, L3/CL3 is a cubic homogeneous relation composed of specified products of chord lengths and side lengths of the triangle of intersection points. The short proof by Chamberland and Zeilberger explicitly attributes this result to Jane McDougall.

8.5 Ceva's Theorem for Chords

Let A,B,C,D,E,F be six points arranged in order on a circle. The three chords AD, BE, and CF are concurrent if and only if AB·CD·EF = BC·DE·FA. For the three chords A_1A_4, A_2A_5, and A_3A_6 considered in this paper, this relation characterizes their concurrency. If the three chords are concurrent, then $P_1=P_2=P_3$, the triangle of intersection points degenerates, and every distance between intersection points appearing in L3 and CL3 is zero. This paper primarily studies the nondegenerate configuration in which the three intersection points are distinct. Thus, Ceva's theorem for chords should be compared as a concurrency criterion and as a description of the degenerate boundary case, rather than identified with L3 itself.

8.6 Summary of the Comparison with the Five Classical Results

The intersecting chords theorem gives a product relation among segment lengths when two chords intersect at one point; Ptolemy's theorem relates the sides and diagonals of a cyclic quadrilateral; Fuhrmann's theorem gives a cubic side–diagonal relation for a cyclic hexagon; the McDougall relation applies to any even number of points on a circle and uses an alternating sum of reciprocals of global distance products; and Ceva's theorem for chords characterizes the concurrency of three alternately paired chords in a six-point circle. The central object of L3/CL3 is the triangle formed by the pairwise intersections of three chords. Table 1 summarizes the differences in objects and forms of expression.

9. Computational Verification and Reproducibility

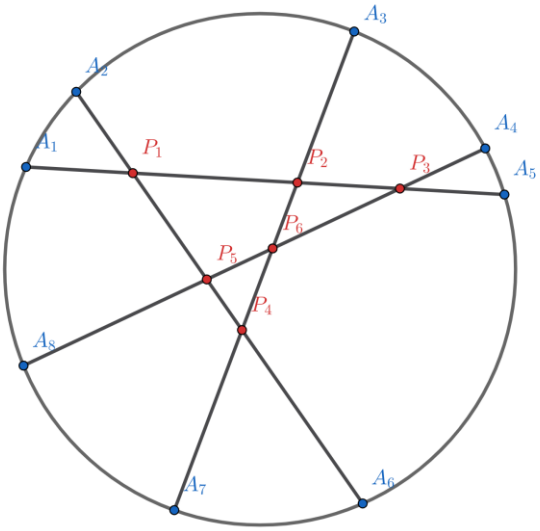
To check consistency among the labels, transcribed formulas, and implementation, an independent Cartesian-coordinate implementation was used for randomized floating-point verification. Six angles were generated randomly on the unit circle and sorted; the three line intersections were computed from their endpoint coordinates; chord lengths and distances between intersection points were calculated directly; and the relative differences between the two sides of L3 and CL3 were then evaluated. The random test used the fixed seed 20261024, with 10,000 accepted samples.

In this computation, the maximum relative difference for L3 was 8.758e-12 and the median relative difference was 2.001e-16; for CL3, the maximum relative difference was 5.114e-12 and the median relative difference was 1.478e-16. The floating-point results are consistent with the scale of

double-precision rounding error. This test can detect common indexing or sign errors, but a finite random sample cannot replace the analytic proofs in Sections 4 and 6, nor does it establish originality.

The verification procedure can be reproduced as follows: (1) fix the random seed; (2) generate and sort six angles on the circle; (3) compute line intersections using two-dimensional vector cross products; (4) calculate all distances independently; (5) evaluate the relative residuals of L3 and CL3 separately; and (6) exclude samples in which near-parallelism or near-degeneracy makes the denominator extremely small, while recording the number excluded. To strengthen reproducibility further, future work may add high-precision computation, exact parameterization, and stress tests for configurations close to degeneracy.

10. Generalization to L4 and General Ln



10.1 L4: Four Chords

In general position, four chords intersect pairwise, producing C(4,2)=6 intersection points. Unlike L3, these six points do not naturally form just one triangle, so a generalization cannot be obtained merely by changing the subscript 6 to 8. First, the pairing pattern of the four chords and the names of the six intersection points must be fixed. Next, one must choose the permitted graph of distances between intersection points (that is, which pairs of intersection points are connected). Candidate monomials of the form “chord length × chord length × distance between intersection points” can then be selected according to cubic homogeneity, with cyclic, reflection, or other symmetry constraints imposed.

It is recommended that candidate identities first be screened by computational search: randomly generate eight points on a circle and pair them into four chords, then compute the six intersections; search the space of candidate monomials for numerical linear relations that are approximately zero;

next perform exact elimination using a rational parameterization of the circle; and finally convert the algebraic relation into a readable geometric proof. If no identity with the prescribed structure exists, research may turn instead to inequalities with explicit assumptions, directions, and equality conditions.

10.2 General Ln

For N chords, the number of pairwise intersections in general position is $C(N,2)=N(N-1)/2$. The number of candidate distances and products of chord lengths grows rapidly, so the target structure should be defined before undertaking unrestricted enumeration. A practical research workflow is: fix the configuration and labels → enumerate homogeneous monomials → perform random screening → use exact rational parameterization → carry out symbolic elimination and factorization → extract the proof structure → search Chinese- and English-language literature.

An important technical issue is the cyclic action: for a configuration of N chords, a cyclic shift acts simultaneously on endpoint pairings, intersection labels, and distances between intersection points. As in the proof of CL3, the generalization must list the mapping of local intersection points to avoid mistaking local symbols for fixed geometric points. L3/CL3 can serve as a model for the research workflow and notation audit, but they do not automatically guarantee the existence of an identity of the same type for arbitrary N.

11. Conclusion

This paper has provided an analytic proof of L3 and discussed CL3 through six cyclic shifts and the mapping of intersection-point labels. Comparison with five classes of classical results shows that the configuration studied here should be distinguished from the single-point intersecting-chords product relation, Ptolemy's relation for quadrilaterals, Fuhrmann's relation for hexagons, McDougall's reciprocal-sum relation for an even number of points, and the concurrency criterion in Ceva's theorem for chords. These comparisons clarify the mathematical background and the boundaries of the configuration; they do not amount to proofs of algebraic independence, irreducibility, or historical priority. Future work includes a more systematic literature review, archiving of exact computations, and investigation of candidate structures for L4/Ln.

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Appendix A. Audit of Key Formulas and Notation

- Fixed definitions: $P_1=A_2A_5\cap A_3A_6$; $P_2=A_1A_4\cap A_3A_6$; $P_3=A_1A_4\cap A_2A_5$.
- The four terms in L3 involve P_1P_3, P_1P_2, P_2P_3, and P_1P_2, respectively. Do not interchange P_1P_2 and P_2P_3.
- In the six-row cyclic summation for CL3, the local Q points are mapped back to the fixed P points according to the table; each coefficient type occurs twice, and the result is finally divided by 2.
- All distances are nonnegative Euclidean lengths. The positive sine branches in the trigonometric parameterization are guaranteed by $a,b,c,d,e>0$ and their sum being less than π.
- Numerical verification is used for error checking; rigorous validity is established by the analytic proofs.